Speculator (§3)

[inputs]

Ensembled Small Speculative Models (SSMs)

Fine-tuned SSM

Quantized LLM

Distillated LLM

Knowledge Retriever

User-Defined Function

Output 0: [machine] learning algorithm
Output 1: [machine] learning system design
Output 2: [machine] translation model

Linearization

$t_{2,0}$ algorithm
$t_{1,0}$ learning
t_0 machine
$t_{2,1}$ system → $t_{3,0}$ design
$t_{1,1}$ translation → $t_{2,2}$ model

Speculated Token Tree

Verified output: machine learning system optimization

Synchronization

$t_{2,0}$ algorithm → $t_{3,1}$ design
$t_{1,0}$ learning
$t_{3,0}$ design → $t_{4,0}$ [EOS]
t_0 machine
$t_{2,1}$ system
$t_{3,2}$ optimization
$t_{1,1}$ translation
$t_{2,2}$ model → $t_{3,3}$ [EOS]

Verified Subtree

Verifier (§4)

Linearization

[machine] → learning → algorithm system → design translation → model

Parallel Decoding

LLM

learning system design optimization [EOS] model [EOS]